

CARLADrive: A Synthetic Multimodal Perception Dataset for Autonomous Driving

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Abstract—Advances in autonomous driving heavily rely on large-scale multimodal datasets for training and evaluating perception models. Existing real-world benchmarks, such as KITTI, nuScenes, and Waymo, drive progress but remain limited by high annotation costs, sensor synchronization issues, and scarce coverage of safety-critical scenarios. Synthetic datasets offer scalability and controllability, yet often lack long temporally coherent sequences, radar sensing, or realistic sensor setups. We introduce CARLADrive, a large-scale synthetic dataset built on CARLA Leaderboard 2.0 routes featuring a nuScenes-inspired sensor suite: six RGB cameras, one LiDAR, and five radars, complemented by a semantic LiDAR and an instance segmentation camera. Data is collected using PDM-Lite, a privileged rule-based agent capable of reliably completing complex urban routes. CARLADrive provides over 1.6M annotated 3D bounding boxes within a 60-meter perception range, along with ego dynamics, future trajectories, and navigation commands. We establish strong baselines using LiDAR, camera, and fusion-based detectors from MMDetection3D, and evaluate a radar-aware model (CaR1), offering a comprehensive benchmark to advance multimodal perception and sensor fusion. Dataset and code will be released upon acceptance. Code and dataset are available online.

I. INTRODUCTION

Autonomous driving perception relies heavily on large-scale, diverse datasets. Real-world datasets like KITTI [1], nuScenes [2], and Waymo Open Dataset [3] present significant limitations regarding annotation costs and the coverage of safety-critical corner cases. Synthetic alternatives generated using CARLA enable controlled experimentation of dangerous driving situations. However, existing synthetic datasets focus on limited perception tasks (CarlaScenes [4]) or short scenarios featuring unpredictable agent behavior (Bench2Drive [5]). Few provide the rich multimodal sensor setups and privileged annotations required for high-fidelity 3D scene understanding.

To address these limitations, we introduce CARLADrive, a large-scale synthetic dataset built upon CARLA Leaderboard 2.0 routes. Our dataset features a nuScenes-inspired sensor

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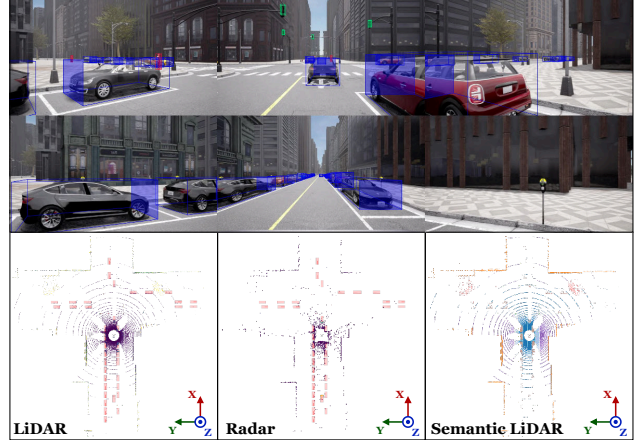


Fig. 1. A sample from the CARLADrive dataset. The figure displays the six RGB camera views, the BEV LiDAR point cloud, the aggregated BEV point cloud from five radars, and the semantic LiDAR point cloud in BEV.

configuration providing synchronized multimodal observations. Data collection is driven by PDM-Lite [6], a privileged rule-based agent capable of completing complex urban routes reliably. CARLADrive includes rich ground-truth annotations (3D bounding boxes, future trajectories, vehicle intentions) and incorporates a semantic LiDAR alongside a front-view instance segmentation camera.

Our main **contributions** are:

- A new multimodal dataset for perception, collected across CARLA Leaderboard 2.0 routes using the PDM-Lite agent.
- A sensor suite of 6 cameras, 1 LiDAR, and 5 radars with precise temporal synchronization, complemented by a semantic LiDAR and an instance-level segmentation camera (Fig. 1).
- Rich ground-truth annotations including 3D bounding boxes, actor trajectories, and vehicle intention labels covering diverse weather and conditions (Fig. 2).
- Extensive benchmarking using SOTA perception models from MMDetection3D [7] (LiDAR, camera, and fusion) and a radar-aware baseline (CaR1 [8]).

We believe this dataset will serve as a valuable benchmark for researchers working on multimodal perception, sensor fusion, and robust 3D scene understanding in autonomous driving systems. We also release the code for data generation, enabling researchers to add, modify, or customize the sensor suite, routes and data collector as desired.

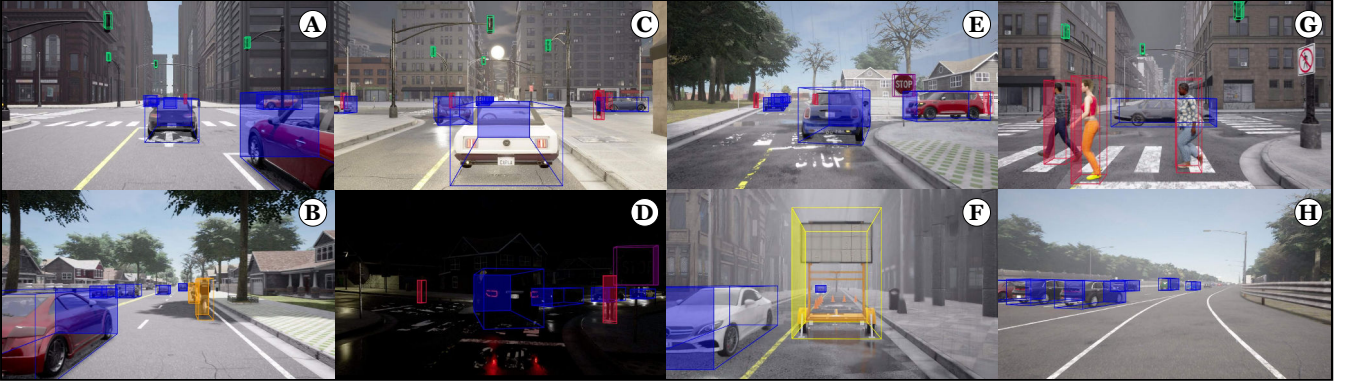


Fig. 2. Front camera samples under diverse conditions. (A–B) clear weather during daytime, (C) dusk, (D) nighttime, (E) light rain, (F) heavy rain, (G) light rain with pedestrian crossing, and (H) clear daytime highway cut-in. Every class in the dataset is represented in the composition: car (blue), walker (red), bicycle (orange), traffic light (green), stop sign (purple), and static traffic warning (yellow).

II. RELATED DATASETS

Real-world datasets (KITTI [1], nuScenes [2], Waymo Open Dataset [3]) face practical limitations regarding acquisition costs and control over environmental diversity. Early synthetic datasets like SYNTHIA [9] and Virtual KITTI 2 [10] lack the multimodal synchronization required for full-stack perception.

A. Synthetic Datasets in CARLA

Recent datasets leveraging CARLA include CarlaScenes [4], Bench2Drive [5], and LangAuto [11]. CarlaScenes focuses on odometry with a single front-facing camera, while LangAuto is optimized for instruction following without radar integration. Bench2Drive is the most comparable alternative, offering a multimodal sensor suite. However, it targets end-to-end driving evaluations on short 150-meter clips. In contrast, CARLADrive focuses on continuous, long-tailed modular perception, providing sequence lengths averaging 8.7 km to enable long-horizon behavior prediction and consistent trajectory-level analysis.

B. Privileged Agents

Synthetic data collection requires privileged agents with access to the simulation’s ground-truth state. Bench2Drive utilizes Think2Drive [12], trained via reinforcement learning. Its stochastic behavior introduces variability that affects dataset consistency, particularly in long-horizon tasks. CARLADrive employs PDM-Lite, a deterministic rule-based expert. This design guarantees the consistent trajectories and stable semantic behavior essential for tasks involving temporal modeling and sensor fusion over extended trajectories. Furthermore, our dataset uniquely includes rare modalities such as semantically annotated LiDAR point clouds and actor-level intention labels.

III. THE DATASET

This section will offer information about the insights of the dataset, regarding the full sensor configuration, the variety of routes and scenarios, the types of annotations and an explanation on the used privileged-agent.

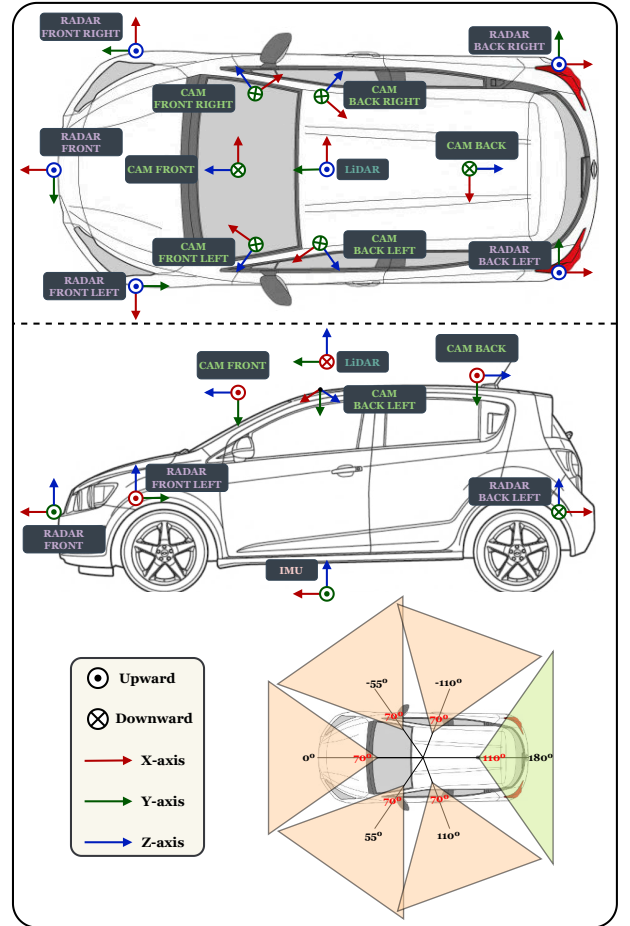


Fig. 3. Sensor setup of the simulated ego vehicle, showing the arrangement of cameras, LiDAR, and radar sensors used for multimodal data collection.

A. Sensor Suite and configuration

Our sensor setup is heavily inspired by the nuScenes dataset, aiming to replicate a realistic perception system. As detailed in Table I and Fig. 3, the front and side cameras

TABLE I
SENSOR CONFIGURATION SUMMARY OF CARLADrive. HEAVILY BASED ON NUSCENES.

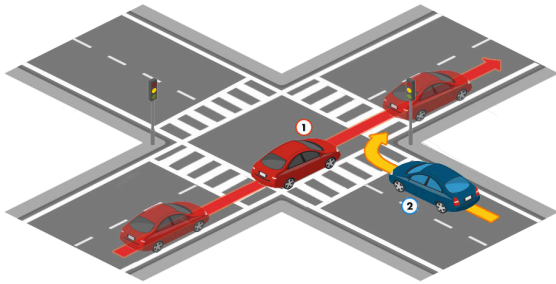
Sensor	Specifications	Purpose
6x RGB Cameras	1600×900 px, JPG format	360° visual coverage
1x LiDAR	64 channels, 10 Hz, 80 m range, 600k points per second 40° Fov (+10° to -30°)	3D geometry and depth
5x radars	80 m range, $\pm 60^\circ$ HFoV, $\pm 14^\circ$ VFoV, 10k points per second	Adverse conditions, occlusion robustness, velocity
1x IMU & GNSS	Location, yaw, speed, acceleration, angular velocity	Ego-motion and localization
1x Instance Seg. Camera	1600×900 px, pixel-level instance masks	2D instance-level supervision
1x Semantic LiDAR	Same specifications as LiDAR, semantic class per point	Point cloud segmentation

possess a horizontal FoV of 70° with an angular offset of 55° , while the rear camera utilizes a 110° FoV to ensure full 360° coverage. To emulate natural LiDAR point loss from occlusions and surface reflectivity, we apply a simulated dropoff rate of 0.45.

Data is synchronized at each simulation tick and stored at an effective sampling rate of 2 Hz, matching nuScenes’ acquisition rate. While other datasets like Bench2Drive sample faster (10 Hz), our focus remains on continuous, long-tailed perception over substantially longer route sequences. Furthermore, we enhance the dataset’s downstream utility by including a semantic LiDAR for per-point semantic labels and an instance segmentation camera for high-precision 2D bounding boxes.

B. Scenarios

CARLADrive is built upon the CARLA Leaderboard 2.0 benchmark. Instead of random data collection, we gather data across diverse challenging conditions including urban downtowns, neighborhoods, and highways. The Leaderboard defines 90 training and 20 validation routes. Along these continuous sequences, agents encounter complex NHTSA-topology traffic scenarios such as unprotected turns, merging, and yielding to emergency vehicles (Fig. 4). This exposes perception systems to realistic occlusions and interactions.



Right turn at an intersection with crossing traffic

Fig. 4. Example scenario from the CARLA Leaderboard Scenarios [13]. The ego vehicle is shown in blue with its trajectory indicated in orange, while the adversarial vehicle is depicted in red.

C. PDM-Lite agent

The choice of the privileged agent heavily impacts dataset quality. We employ PDM-Lite, a rule-based expert de-

signed for urban driving. Unlike learned agents that exhibit stochastic behavior (which introduces unwanted variability in long-horizon tasks), PDM-Lite operates deterministically using predefined rules derived from complete environmental knowledge. This guarantees robust, repeatable behavior and consistent trajectories, which recent evaluations show yields nearly twice the success rate in complex scenarios compared to reinforcement learning agents.

D. Dataset statistics

The dataset provides large-scale, diverse, and realistic multimodal data collected using the CARLA simulator under varied driving conditions. Given our data gathering conditions, we release 2 different splits for the dataset:

- **Mini:** This split includes 4 training routes and 1 validation route, designed for quick prototyping and small-scale experiments. It provides approximately 12,000 training samples and 4,000 validation samples, requiring around 80 GB of storage.
- **Standard:** This split comprises 30 training routes and 5 validation routes, serving as the main experimental setting for this paper. It contains about **82,000** training samples and **27,000** validation samples, roughly three times the size of nuScenes, occupying around 400 GB. It spans over **260 km** of driving and more than **11 hours** of data collection across **30** distinct urban routes, all aligned with the CARLA Leaderboard 2.0 benchmark. The route lengths and durations are shown in Figure 11.

To ensure robustness, data spans multiple clear, cloudy, and rainy conditions, significantly increasing weather variability compared to nuScenes (Fig. 5). Traffic density and interaction complexity are captured in Fig. 6 and Fig. 7.

Crucially, CARLADrive provides over 1.6 million 3D bounding boxes within a 60-meter radius. Beyond standard dynamic classes (Fig. 8, Fig. 9, Fig. 10), we explicitly annotate traffic lights and stop signs—elements critical to road environments yet rarely annotated in real-world datasets due to labeling complexity. The dataset is rounded out with dense LiDAR semantic point clouds (Fig. 12) and high-level navigation commands, intention labels, and ego-velocities to support behavior prediction and trajectory forecasting.

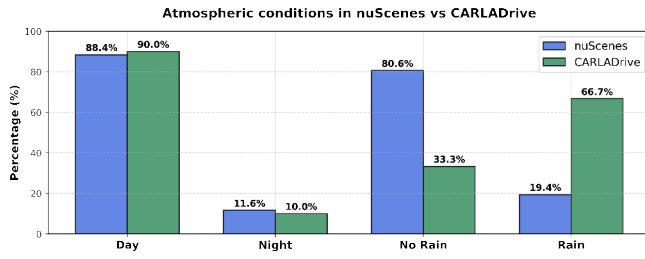


Fig. 5. Comparison of dataset conditions. CARLADrive separates rain into *soft* and *hard* categories, corresponding to CARLA precipitation levels below or above 50%.

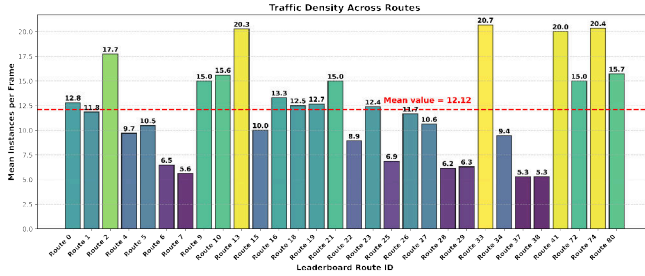


Fig. 6. Traffic density across routes. Bars indicate the mean number of instances per route, with the overall average of 12.39 shown as a dashed line.

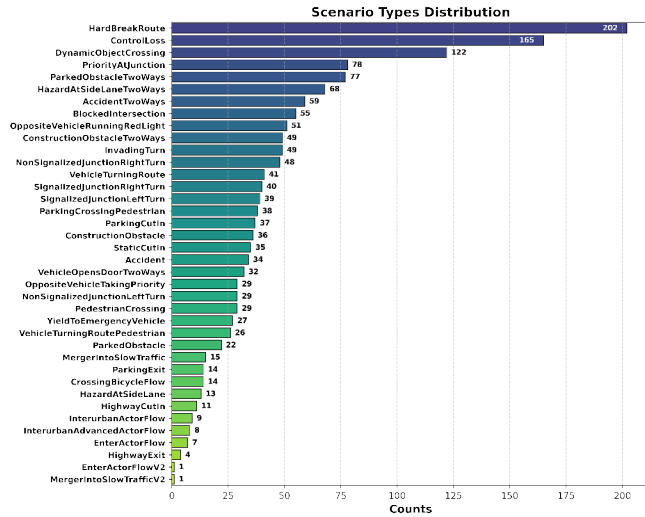


Fig. 7. Distribution of CARLA Leaderboard scenarios across the dataset.

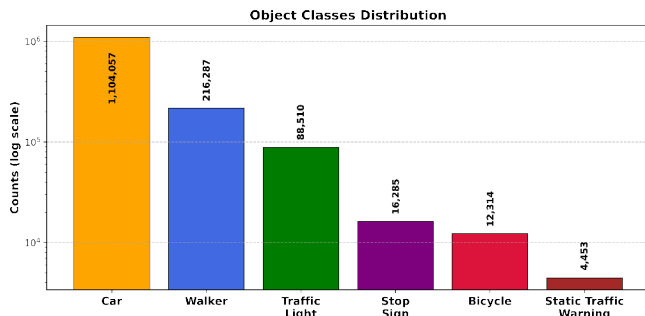


Fig. 8. Distribution of object classes in the training and validation splits. Annotated classes are: *car*, *walker*, *traffic light*, *stop sign*, *bicycle*, and *static traffic warning*.

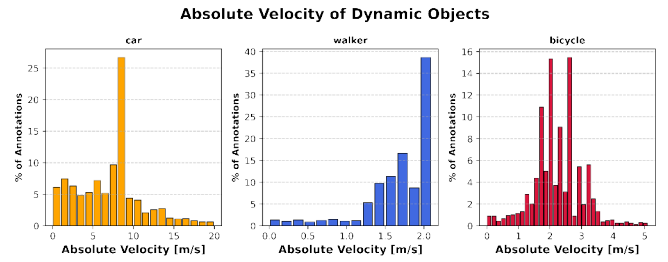


Fig. 9. Absolute velocity distribution of dynamic objects in the training split. Only objects with *speed* > 0.05, m/s are considered.

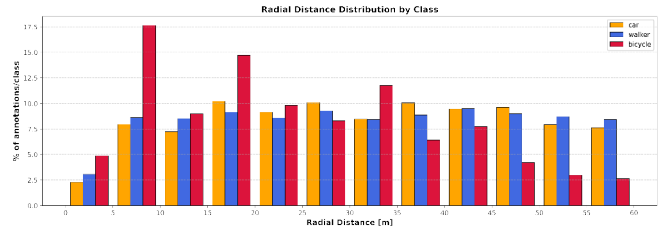


Fig. 10. Radial distance distribution of *car*, *walker* and *bicycle* classes.

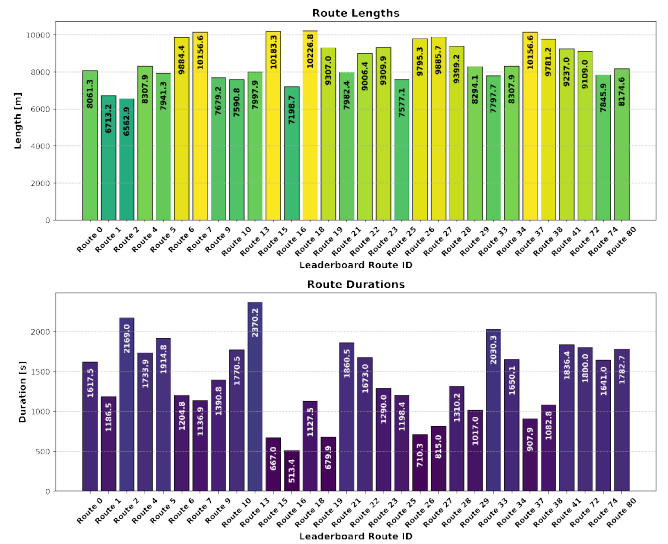


Fig. 11. Train routes lengths [m] and durations [s]. Routes with highway segments cover longer distances in less time. Sample count per route is twice the duration (2 Hz).

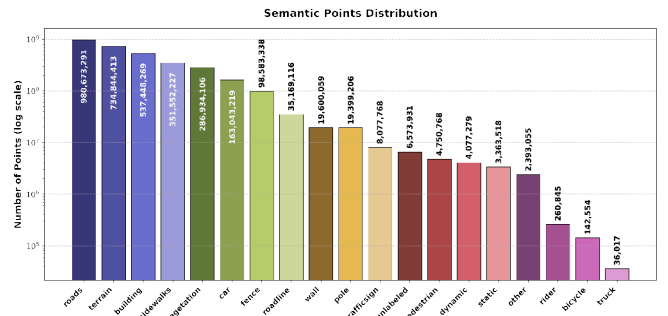


Fig. 12. Semantic LiDAR points distribution. Semantic classes are: *roads*, *sidewalks*, *building*, *wall*, *fence*, *pole*, *traffic sign*, *vegetation*, *terrain*, *pedestrian*, *rider*, *car*, *truck*, *bicycle*, *static*, *dynamic*, *other*, and *roadline*.

IV. EXPERIMENTS AND BASELINES

A. MMDetection3D

To establish consistent and comparable baselines across different perception tasks with CARLADrive, we leverage **MMDetection3D** [7], an open-source, modular 3D object detection framework built on top of the popular MMDetection toolbox [18]. MMDetection3D provides unified implementations for state-of-the-art models covering camera-based, LiDAR-based, and multimodal 3D detection, including both anchor-based and anchor-free paradigms. Its standardized codebase, dataset handling tools, and robust evaluations pipelines allow for fair comparisons across different sensor modalities and fusion strategies.

Moreover, MMDetection3D includes support for widely-used benchmarks such as KITTI, nuScenes, and Waymo. To ensure seamless compatibility, we adapt and extend the framework to fully integrate our dataset, enabling direct training, evaluation, and comparison of models within the same experimental environment. This guarantees reproducibility, scalability, and alignment with standard experimental protocols, while also facilitating the deployment of cutting-edge 3D perception models on synthetic yet realistic multimodal data.

To establish strong detection baselines on our dataset, we evaluate a set of SOTA models covering different sensor modalities and architectures, as summarized in Table II.

B. CaR1

CaR1 [8] is a framework for Bird’s Eye View (BEV) semantic segmentation that explores multimodal fusion of cameras, radar, and LiDAR in a unified representation space. Unlike conventional 3D object detectors that predict discrete bounding boxes, CaR1 generates dense, pixel-level semantic maps in the BEV plane, providing detailed scene understanding and spatial awareness for downstream perception and planning tasks.

The architecture combines a backwards interpolation view transformer for multi-camera image features and a lightweight Point Transformer as radar encoder for processing radar point clouds. Additionally, we add a LiDAR branch based on point pillarization. Sensor fusion is performed following BEVFusion principles through an adaptive convolutional mechanism with dynamic sensor weighting. An efficient Attention U-Net decoder then produces dense BEV semantic masks, focused on road users and obstacles.

In this work, we evaluate CaR1 across several sensor configurations: Camera-only (Ca), Radar-only (R), LiDAR-only (L), Camera + Radar (CaR1) and Camera + LiDAR (CaL) fusion setups. This allows for a comprehensive comparison of sensor contributions in BEV segmentation tasks, complementing the detection-focused benchmarks provided by MMDetection3D models.

C. Results

To demonstrate the potential of CARLADrive, this paper presents results on two key perception tasks: 3D object detection and BEV segmentation. 3D object detection results

are presented in Table II, while BEV segmentation results across all sensor configurations are summarized in Table III.

Evaluation metrics differ between tasks. For 3D object detection models implemented using MMDetection3D, we report the mean Average Precision (mAP) with an Intersection over Union (IoU) threshold of 0.7 for the *car* class and 0.5 for all other classes, following the MMDetection3D evaluation protocol:

$$\text{mAP} = \frac{1}{|C|} \sum_{c \in C} \text{AP}(c) \quad (1)$$

where C is the set of classes, $|C|$ is the number of classes, and $\text{AP}(c)$ denotes the Average Precision for class c at the specified IoU threshold.

For BEV segmentation with CaR1, we use Intersection over Union (IoU), also known as the Jaccard Index, computed only for the *car* class:

$$\text{IoU} = \frac{B_p \cap B_{gt}}{B_p \cup B_{gt}} \quad (2)$$

where B_p represents the set of predicted pixels in the BEV map, and B_{gt} represents the set of ground-truth pixels.

For all experiments, instances that do not receive LiDAR point hits are excluded from processing. Nevertheless, these instances remain in the dataset labels for users. All MMDetection3D models were trained for 6 epochs due to the large dataset size, while CaR models were trained for 10 epochs.

The results in Tables II and III reveal consistent patterns. LiDAR-based methods achieve the highest overall performance, confirming their robustness and reliability for 3D object detection under varied conditions. Camera-only models deliver competitive accuracy but generally underperform compared to point-based approaches. Fusion strategies consistently strengthen robustness by leveraging complementary sensor information, leading to superior results over single-modality baselines. Notably, the CaR1 configuration demonstrates that radar-aware models significantly outperform camera-only systems and approach LiDAR-only performance, despite radar traditionally being weaker in object-level perception.

A closer analysis of the class-wise metrics highlights specific challenges. The most difficult categories are *traffic light* and *stop sign*, where camera-only detectors struggle significantly. Incorporating LiDAR within BEVFusion improves *stop sign* detection, but not *traffic lights*, since very few LiDAR points are captured on such small and elevated targets. The *walker* class is also demanding for simpler architectures, though advanced models such as CenterFormer exceed 20% AP. In contrast, road users like *bicycle* and especially *car* are detected more reliably, with cars reaching nearly 50% AP in SOTA methods. Finally, *static traffic warning* emerges as a highly distinctive class, where camera-based models already perform strongly.

Overall, these findings validate CARLADrive as a suitable benchmark for multimodal perception and underscore the important contribution of radar sensing to enhancing safety in autonomous driving.

TABLE II
3D OBJECT DETECTION RESULTS FROM MMDetection3D.

Model	Modality	walker	bicycle	car	traffic light	stop sign	static traffic warning	mAP	FPS
DETR3D [14]	Camera	9.13	25.10	35.60	1.37	1.81	75.40	24.74	6.88
PointPillars [15]	LiDAR	9.10	19.45	42.98	—	—	—	23.84	313.97
CenterFormer [16]	LiDAR	21.44	20.20	49.92	—	—	—	30.52	28.62
BEVFusion [17]	Camera + LiDAR	25.51	24.37	42.08	0.23	38.99	82.06	35.54	16.70

For point-based detectors, only dynamic categories are evaluated (*walker*, *bicycle*, *car*). Inference speed (FPS) is reported on an NVIDIA GeForce RTX 4090. Best is marked in **bold**.

TABLE III
BEV SEGMENTATION RESULTS FROM CARL.

Modality	FPS	IoU
Camera-only (Ca)	16.96	52.04
Radar-only (R)	17.79	54.45
LiDAR-only (L)	18.60	71.14
Camera + Radar (CaR)	11.80	62.37
Camera + LiDAR (CaL)	10.71	77.61

Only *car* class is evaluated. Inference speed (FPS) is reported on an NVIDIA GeForce RTX 4090. Best is marked in **bold**.

V. CONCLUSIONS AND FUTURE WORKS

In this work, we introduce CARLADrive, a large-scale synthetic multimodal dataset for autonomous driving, generated along CARLA Leaderboard 2.0 routes with a nuScenes-inspired sensor suite. It provides over 1.6 million annotated 3D bounding boxes within a 60-meter perception range, along with ego-vehicle dynamics, future trajectories, and high-level navigation commands. By combining multiple RGB cameras, LiDAR, radars, and semantic annotations, CARLADrive enables development and evaluation of perception models under diverse urban scenarios and weather conditions. We also establish baselines using state-of-the-art detectors from MMDetection3D and a radar-aware model (CaR1), demonstrating the dataset’s suitability for benchmarking multimodal perception and sensor fusion approaches.

This release constitutes the first version of the CARLADrive dataset, and we plan to expand and refine it over time. The next step aims to include all 90 training routes and 20 validation routes from the CARLA Leaderboard, resulting in a dataset roughly three times larger than the current version. Moreover, datasets of this kind enable the establishment of baselines not only for 3D perception tasks but also for higher-level challenges, including instance prediction, trajectory forecasting, and even end-to-end autonomous driving tasks such as control action or waypoint generation. Looking ahead, we also plan to provide ground-truth annotations for 3D occupancy prediction, following the guidelines established by Occ3D [19].

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